

Deep Learning Image Captioning in Construction Management: A Feasibility Study

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Abstract: Deep learning image captioning methods are able to generate one or several natural sentences to describe the contents of construction images. By deconstructing these sentences, the construction object and activity information can be retrieved integrally for automated scene analysis. However, the feasibility of deep learning image captioning in construction remains unclear. To fill this gap, this research investigates the feasibility of deep learning image captioning methods in construction management. First, a linguistic schema for annotating construction machine images was established, and a captioning data set was developed. Then, six deep learning image captioning methods from the computer vision community were selected and tested on the construction captioning data set. In the sentence-level evaluation, the transformer-self-critical sequence training (Tsfm-SCST) method has obtained the best performance among six methods with the bilingual evaluation (BLEU)-1 score of 0.606, BLEU-2 of 0.506, BLEU-3 of 0.427, BLEU-4 of 0.349, metric for evaluation of translation with explicit ordering (METEOR) of 0.287, recall-oriented understudy for gisting evaluation (ROUGE) of 0.585, consensus-based image description evaluation (CIDER) of 1.715, and semantic propositional image caption evaluation (SPICE) score of 0.422. In the element-level evaluation, the Tsfm-SCST method achieved an average precision of 91.1%, recall of 83.3%, and an F1 score of 86.6% for recognition of construction machine objects by deconstructing the generated sentences. This research indicates that deep learning image captioning is feasible as a method of generating accurate and precise text descriptions from construction images, with potential applications in construction scene analysis and image documentation. DOI: [10.1061/\(ASCE\)CO.1943-7862.0002297](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002297). © 2022 American Society of Civil Engineers.

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Introduction

Construction videos contain important visual information (e.g., working objects and their activities) that is of benefit for project management purposes (Xiao and Zhu 2018). By analyzing construction videos using vision-based methods, many applications can be developed to automatically monitor crew productivity, identify safety risks, and optimize working spaces (Yang et al. 2015). For example, Chen et al. (2020) developed a vision-based system to calculate excavator productivity in earthmoving. Kolar et al. (2018) proposed a vision-based method for identifying safety guardrails at construction sites in order to prevent workers from accessing hazardous site areas. Compared with other advanced technologies (e.g., laser scanners, radio frequency identification, and global positioning systems), vision-based monitoring of construction sites offers the advantages of lower cost, simple deployment, and easy maintenance (Xiao and Kang 2021b).

Scene analysis refers to interpreting the construction image by identifying construction elements (e.g., objects, activities, and relationships between objects), which provides a holistic view of visual information pertaining in images to construction

management (Liu et al. 2020). The applications of scene analysis include automated detection of safety rule violation (Wang et al. 2019), roadway asset evaluation (Balali and Golparvar-Fard 2015), and construction hazard identification (Fang et al. 2020). However, existing scene analysis methods are template-based, which refers to retrieving construction elements separately by object detection (Xiao et al. 2021b) or activity recognition (Golparvar-Fard et al. 2013) and combining all information into a predefined template, and these methods have two limitations: (1) the different scene information in construction images is retrieved separately, making it a time-consuming process; and (2) the retrieved scene information is usually combined based on predefined orders to generate a set of words or a sentence as the final output, but these results are prone to error or incompleteness.

Image captioning refers to the automatic generation of one or several sentences to describe the contents of an image; it is a disciplinary technology rooted in computer vision and natural language processing (Hossain et al. 2019), which can be potentially used for construction scene analysis. Recently, deep learning methods have enjoyed considerable success because they are capable of extracting high-level features automatically from images for various applications in computer vision, natural language processing, reinforcement learning, and so on (Lecun et al. 2015). By incorporating deep learning, image captioning methods can generate precise and concise text descriptions from images by training on annotated image data sets. A typical deep learning image captioning method is the encoder-decoder architecture (Mao et al. 2014), where a convolutional neural network (CNN) encoder extracts embedding features from the images, and a recurrent neural network (RNN) decoder generates texts based on the embedded features. By adopting deep learning image captioning, the scene information in construction images or videos can be retrieved integrally in the form of a natural sentence.

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Although deep learning image captioning has achieved considerable success within the computer vision community, few studies have adopted these methods in construction scenarios, and the feasibility of deep learning image captioning has yet to be substantiated in the following respects: (1) a method of integrating deep learning image captioning methods for generating construction text descriptions is not available—most studies of deep learning image captioning are conducted in daily life scenarios; (2) a linguistic schema for annotating construction images has yet to be established, and an annotated data set of construction images is needed in order to train deep learning image captioning algorithms for construction management purposes; and (3) studies within the construction research domain have typically employed only the basic CNN-RNN method, and advanced deep learning methods (e.g., attention) have not yet been tested in construction applications.

The overall objective of the research described herein is to investigate the feasibility of deep learning image captioning in construction scenarios. To achieve this goal, we propose a method for integrating deep learning image captioning methods in construction scenarios that consists of three main steps: data set development, model establishment, and experimental evaluation. The results of this research demonstrate the practical applicability of deep learning image captioning and underscore its potential in the context of construction management. Currently, studies of deep learning image captioning are still in the early stage in the field of construction management. The results of this research are expected to help researchers in our field to decide if and how to use deep learning image captioning methods in their future construction studies.

Literature Review

Automatic analysis of construction images or videos using vision-based methods is an important emerging avenue of construction research with potential project management benefits in terms of increasing crew productivity, reducing safety risks, and monitoring project progress. In this section, existing studies of applying vision-based methods in construction sites are firstly reviewed in terms of methods and applications. Then, relevant works on deep learning image captioning are reviewed in a comprehensive way. Finally, the research gaps and the objectives of the present study are identified.

Vision-Based Monitoring in Construction

Existing vision-based methods adopted in construction include object detection, object tracking, activity recognition, and scene analysis. Many construction applications (Roberts and Golparvar-Fard 2019; Son et al. 2019; Wang and Zhu 2021) have been developed based upon these four types of vision-based methods. Object-detection methods retrieve the localization and categorical information of construction objects (e.g., machines, workers, and materials) from construction images or videos (Kulchandani and Dangarwala 2015), which is the fundamental step for many vision-based studies in construction. For instance, Yin et al. (2020) adopted the deep learning detection method You Only Look Once (YOLO)-v3 (Redmon and Farhadi 2018) to detect defects in sewer pipes. Kim et al. (2018) proposed a region-based fully convolutional network for detecting construction machines that achieved a mean average precision of 96.33% in their experiments.

Object tracking aims to retrieve and interpret the movements of construction objects in continuous frames, which produces the trajectory information by assigning an identification number (ID) to each object. For example, Zhu et al. (2017) integrated object detection for the purpose of tracking construction workers and

machines from jobsite videos. Konstantinou et al. (2019) developed a vision-based method for tracking construction workers by integrating an appearance model with a filtering model. Xiao et al. (2021a) adopted a deep learning illumination-enhancement method for tracking multiple construction machines in a nighttime environment.

Activity recognition has been applied in construction monitoring to identify the activities of construction workers and machines in videos, a task that has traditionally relied upon the results of object recognition and object tracking with indicating the activity and localization information of a specific construction object. For example, Yang et al. (2016) proposed a cutting-edge video-interpretation method based on dense trajectories for recognizing the activities of workers. Luo et al. (2018b) adopted the two-stream CNN method for recognizing workers' activities in site surveillance videos, achieving 80.5% accuracy. Furthermore, Chen et al. (2020) have adopted the three-dimensional (3D) ResNet method (Hara et al. 2018) for recognition of excavator activity in earthmoving. Luo et al. (2018a) proposed a two-step method that uses CNN for recognition of construction machine activities from still images from a construction site. What these and other studies available in the literature demonstrate is that activity recognition can be used effectively for advanced vision-based applications including safety control, productivity analysis, and so on.

The goal of scene analysis is to interpret the entire construction image by identifying all construction elements including objects (e.g., machines, workers, and materials), activities of objects, and relationships between objects. Existing scene analysis methods, it should be noted, usually consist of two steps: (1) recognition of construction objects and activities from images by adopting object detection and activity recognition methods, respectively; and (2) identification of relationships between objects based on construction rules. For example, Kim et al. (2016) proposed a data-driven scene analysis method based on a scene dense alignment module for extracting information from construction images. Pour Rahimian et al. (2020) proposed a gamelike hybrid application to simulate real construction activities in a building information modeling (BIM) environment by analyzing construction scenes from videos. Wang et al. (2019) developed a crowdsourcing approach to parse the construction scene from images in order to monitor safety compliance in construction scenarios. So far, an integrated method of automatically generating complete scene information for construction scene analysis, however, has yet to be developed.

Deep Learning Image Captioning

Image captioning, which generates one or several sentences from an image to describe the scene information appearing in the image, is an interdisciplinary research area combining computer vision and natural language processing (Huang et al. 2019). Unlike other vision-based methods (i.e., object detection, object tracking, and activity recognition), the goal of image captioning is to interpret an image by natural sentences from a holistic view instead of providing localization information of construction objects appearing in the image. The image captioning framework generally consists of an encoder and a decoder, where the encoder extracts features from images and the decoder generates texts from the retrieved features (Hossain et al. 2019). Deep learning methods (e.g., CNN and RNN) have performed well in image captioning applications. Mao et al. (2014) proposed a deep learning image captioning method that uses CNN as the encoder and RNN as the decoder; this approach has been built upon in the show-and-tell method by only inputting

the visual features at the first time step of RNN (Vinyals et al. 2017).

Recently, the attention mechanism has been widely applied in deep learning image captioning because it allows the neural network to focus on its subset of inputs in selecting specific features. Xu et al. (2015) introduced the attention mechanism in image captioning, and it has performed well on the Common Objects in Context (COCO) captioning data set (Lin et al. 2014). COCO captioning is a public image captioning data set in the computer vision community, which contains over 1.5 million captions describing over 330,000 images collected from common-life scenarios. Gao et al. (2019), meanwhile, have proposed an adaptive attention approach to image captioning that considers both visual information and language context information for caption generation purposes. Closely related to this, transformer attention (Vaswani et al. 2017), although originally proposed for language translation purposes, was soon adopted in image captioning studies (Vig 2019; Zhang et al. 2019), where it achieved reliable performance. Although the attention mechanism has been successfully applied in the computer vision community, its feasibility in construction scenarios has not yet been investigated.

In construction management, image captioning can be used for automated scene analysis. By analyzing the sentences generated by image captioning methods, the major objects, activities, and interactions of objects can be retrieved integrally, which reduces the processing time and improves the scene analysis completeness. For example, Liu et al. (2020) adopted the CNN and long short-term memory (LSTM) neural networks for captioning construction images to manifest construction worker activity scenes. In their research, a linguistic schema used for annotating images of construction workers was proposed and three experiments were conducted to illustrate the feasibility of image captioning in construction. However, that study only focused on construction workers, and the captioning images of construction machines was not included. Bang and Kim (2020) applied image captioning to drone images for vision-based monitoring of construction sites, achieving a mean average recall (mAP) of 45.52%. However, that study only investigated the use of basic CNN-RNN for image captioning. Although image captioning can provide versatile information for construction scene analysis, it cannot provide effective localization information as to what object detection, object tracking, and activity recognition can offer. Therefore, image captioning can be a promising alternative for construction scene analysis, but it cannot be object detection or activity recognition for vision-based monitoring of construction sites.

Research Gaps and Objectives

As reviewed in the preceding section, deep learning image captioning has enormous potential as a method for construction scene analysis, providing an integrated solution for describing construction objects and their activities as captured in images. The following research gaps have been identified and are addressed in this research:

- A robust method incorporating deep learning image captioning to generate construction text descriptions is currently not available. Such a method can be expected to provide a standard approach for developing a construction image captioning data set, selecting the proper image captioning technique, and evaluating image captioning models for construction management.
- A linguistic schema for annotating construction machine images with professional text descriptions is currently lacking. Such a schema would be capable of bridging the gap between deep learning image captioning and construction management.

Although there exist linguistic schema (Liu et al. 2020) for construction worker images, these cannot be directly applied to construction machine images.

- The performance of deep learning image captioning methods in construction scenarios needs to be investigated. Most studies (Bang and Kim 2020; Liu et al. 2020) in construction only adopted the basic CNN-RNN method for image captioning, whereas the use of advanced techniques (e.g., attention and transformer) in construction has not been validated.

To fill these gaps, this research proposes a feasibility study of deep learning image captioning in construction. To achieve this goal, the following objectives were pursued:

- Propose a method for integrating deep learning image captioning methods in construction scenarios. The proposed method demonstrates a standard approach to applying deep learning image captioning methods to construction scenarios. The present research, although it focuses on construction machine images, can be expanded to images of other construction objects (e.g., workers) in future works.
- Develop a linguistic schema for annotating construction machine images. A linguistic schema instructs the annotator to label construction images with professional descriptions that are precise and accurate and that will be useful for construction management purposes. In this context, deep learning models can be trained to generate professional descriptions for construction scene analysis.
- Compare the performance of the state-of-the-art deep learning image captioning methods built upon different mechanisms in construction scenarios. The comparison results showed the feasibility of deep learning image captioning in construction scenarios. Meanwhile, researchers in the construction community can select the appropriate captioning method for a given application based on the comparative results of this research.

Methodology

Our methodology of integrating deep learning image captioning methods in construction scenarios is introduced in this section. First, the overview of the proposed method is introduced. Then, the details of the data set development and model establishment steps are described.

Methodology Overview

Fig. 1 illustrates the overview of the proposed methodology for incorporating deep learning image captioning methods in construction to generate professional text descriptions from construction images. The method is divided into three main steps: data set development, model establishment, and experimental evaluation. In the data set development step, construction images were first collected from different viewpoints, illumination conditions, and construction stages and projects. Then, the collected images were manually filtered to select qualified images in a manner that ensured the diversity of the data set and mitigated potential overfitting issues. Meanwhile, a linguistic schema for construction machine images is proposed for instructing image annotation. Each machine image was annotated with several sentences to describe what is occurring as captured in the given image. Finally, an image captioning data set was developed in this step.

In the model establishment step, six deep learning image captioning methods were identified based on their reliable performance in the computer vision community. Five evaluation metrics were then selected for automatic evaluation of the deep learning image captioning methods. The implementation of each technique is also

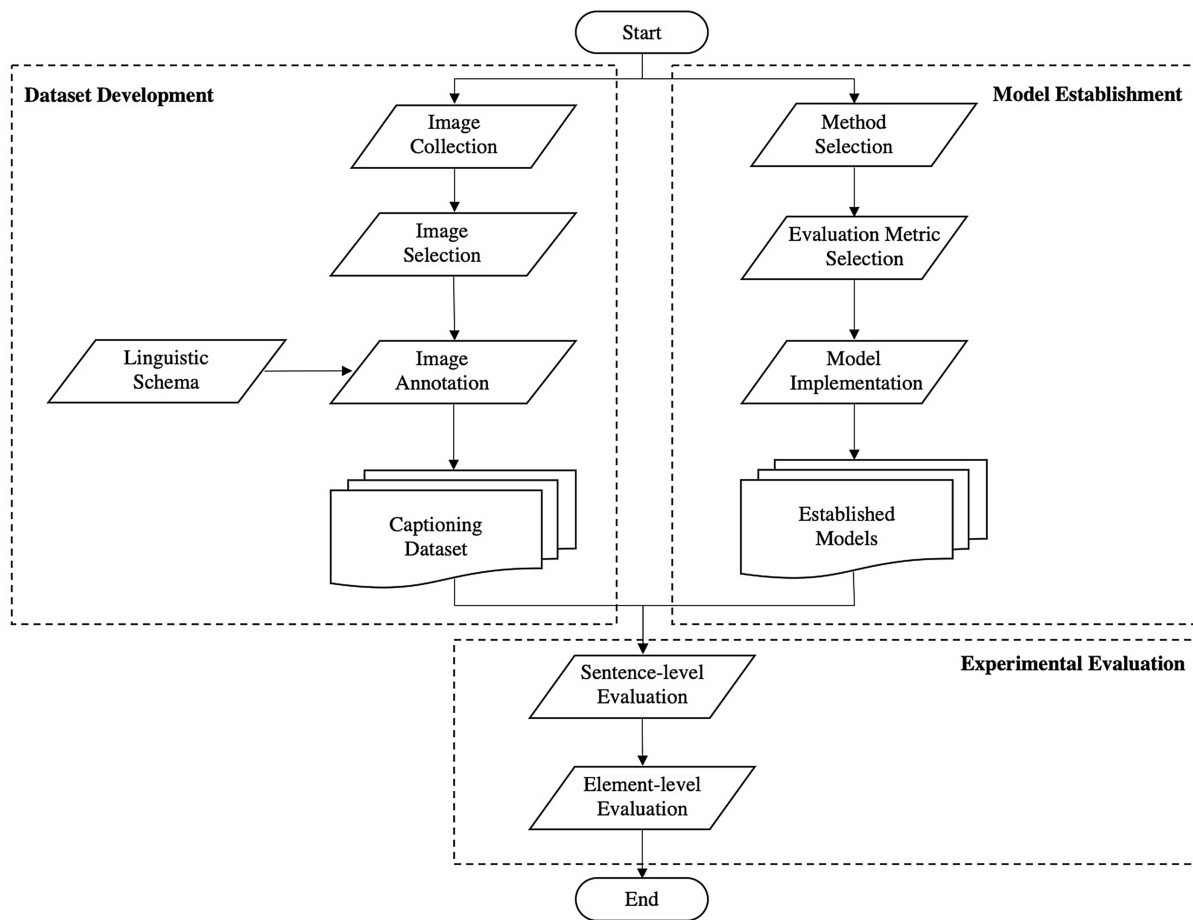


Fig. 1. Overview of the proposed methodology.

demonstrated for model establishment purposes. In the experimental evaluation step, the captioning data set was divided into a training set and a validation set for testing the six image captioning methods. They were compared at both the sentence level and element level in order to validate the feasibility of their use in construction management applications. The data set development and model establishment steps are described in greater detail in the following subsections, and the experimental evaluation and results are described in the next section.

The technical innovations of this study are twofold. First, a standard method is introduced to illustrate how to integrate deep learning image captioning methods in construction, which has three steps including data set development, model establishment, and experimental evaluation. By following this three-step method, other novel image captioning methods can be integrated and evaluated in construction scenarios conveniently. Second, a linguistic schema is proposed to annotate construction machine images with professional text descriptions, which fills the gap of applying deep learning image captioning in construction management. The captioning of other types of construction images (e.g., workers) can be accomplished by following the linguistic schema by updating the object and activity types.

Data Set Development

The development of an annotated image data set is the fundamental step in applying deep learning image captioning in construction management. To develop the data set, three major steps need to be accomplished: image collection, selection, and annotation. In this subsection, the details of image collection and selection and

linguistic schema and image annotation are described, and a comprehensive summary of the captioning data set is provided.

Image Collection and Selection

Collection of construction images with a range of visual characteristics in terms of size, color, shape, and illumination level improves the robustness and generalizability of deep learning image captioning methods in construction scenarios. Xiao and Kang (2021a) developed an image data set of construction machines, referred to as the Alberta Construction Image Data Set (ACID). Images of 10 types of construction machines were included in ACID: excavator, compactor, dozer, grader, dump truck, concrete mixer truck, wheel loader, backhoe loader, tower crane, and mobile crane. The images in ACID were adopted in the present research for further image annotation.

In the authors' previous work (Xiao and Kang 2021a) 124,500 construction images were compiled from online sources (i.e., Google Images, Naver, and YouTube), and 37,500 construction images captured by smartphones, fixed-position cameras, and drones from site visits in Edmonton, Canada, and Xi'an City, China, were also included. Then, the 162,000 images collected were manually processed following four steps: (1) removal of duplicate images; (2) removal of low-resolution images; (3) removal of oversized and undersized images; and (4) blurring of all human faces appearing in the images (for privacy reasons). This process yielded 10,000 construction images that qualified for inclusion in the ACID data set, where Fig. 2 shows some sample images from the ACID data set. In the present research, 4,000 construction images were randomly selected from ACID for the purpose of developing the image captioning data set.



Fig. 2. Example construction images in ACID. (Reprinted from Xiao and Kang 2021a, © ASCE.)

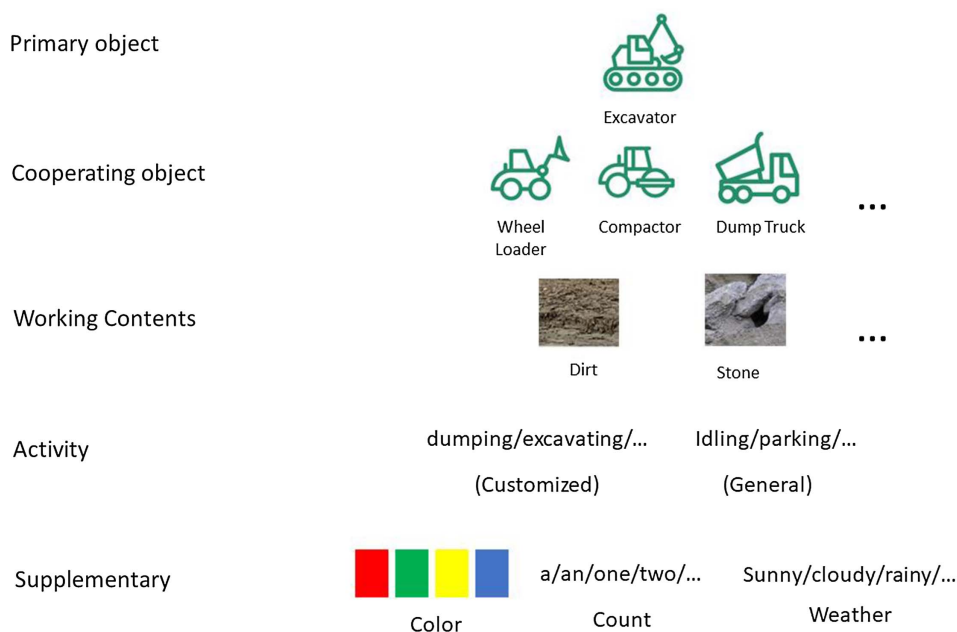


Fig. 3. Illustration of linguistic schema. (Images by authors.)

Linguistic Schema and Image Annotation

For the image captioning data set, each image needs to be manually annotated with several sentences to describe the contents of the image. The linguistic schema informs the process of annotating the construction images, which in turn play an important role in the application of deep learning image captioning in construction. In typical computer vision applications, annotators are required to describe images in their own words because the target images are captured from daily life. In construction, the text annotations of

images must be professional and precise for construction management purposes. As such, the annotators must use specific and correct terms to describe construction objects, activities, and working contents using the linguistic schema rather than simply using their own words as in some other computer vision applications.

Fig. 3 shows the linguistic schema used in this study for annotating construction machine images. First, the following elements must be deconstructed from the construction image according to the linguistic schema: (1) the primary machine object; (2) the machine

Table 1. List of suggested activities for construction machines

Construction machine	Customized activity
Excavator	Swinging/dumping/excavating/loading, and others
Compactor	Compacting and others
Dozer	Grading/stripping/loosening/pushing, and others
Grader	Grading/stripping/loosening/pushing, and others
Dump truck	Dumping/hauling/transferring, and others
Concrete mixer truck	Dumping/transferring/loading, and others
Wheel loader	Dumping/excavating/loading/transferring, and others
Backhoe loader	Dumping/excavating/loading/transferring, and others
Tower crane	Lifting/transferring/swinging, and others
Mobile crane	Lifting/transferring/swinging, and others
General activity	Traveling/waiting/idling/driving/parking, and others

object cooperating with the primary object; (3) the working contents (e.g., dirt, stone, and construction materials) of the primary object; (4) the activities of the primary machine; and (5) supplementary information, such as color, count, and weather conditions. Then, the correct terms must be matched with each element deconstructed in the previous step. Finally, a logical and correct sentence is formed using words to describe what is occurring in the construction image.

Because the construction images under study were drawn from the ACID data set, the primary object and cooperating object terms must be selected from among the 10 designated construction machine types: excavator, compactor, dozer, grader, dump truck, concrete mixer truck, wheel loader, backhoe loader, tower crane, and mobile crane. As a further measure to ensure the accuracy and precision of the annotations, we developed a list of activities (Table 1) for each type of construction machine to serve the activity selection in the linguistic schema, where general activities are those that are applicable to any type of machine. Annotators were encouraged to select the activity of the primary object from the options listed in Table 1, although they were also permitted to use other

activity terms based on their construction knowledge/background if needed.

Thirty volunteer annotators with engineering background from the University of Alberta were recruited to participate in the annotation task. The annotators were first given a half-hour presentation to introduce the research, including an overview of deep learning image captioning, annotation tasks, and the linguistic schema. The volunteers were then assigned a series of construction images and prompted to write one sentence describing the contents of each image, with further instruction and feedback provided by the research in reference to the annotations as needed. The annotation task having been completed by the volunteers, the authors of this research manually checked the annotation results and resolved any errors identified.

Summary of Captioning Data Set

As mentioned previously, 4,000 images from the ACID data set were annotated, and a total of 8,226 captions produced, meaning that each construction image was annotated by two annotators on average. Fig. 4 shows the element distribution in the captioning data set, including the machine terms and activity terms. It can be observed that the excavator and dump truck were the two object terms appearing most frequently in the captioning data set, whereas loading and dumping were the two most frequently used activity terms. Table 2 outlines the top 20 n -grams ($n = 2, 3, 4$) in the captioning data set where the n -gram is defined as the contiguous sequence of n words from natural sentences. As indicated in Table 2, the most frequent terms are found to be “loader is” (2,069), “wheel loader is” (1,151), and “a wheel loader is” (768) for 2-gram, 3-gram, and 4-gram, respectively. The captioning data set was divided into a training set (80%) and a validation set (20%) for the experimental evaluation step.

Model Establishment

Most of the existing image captioning methods in use within the construction domain have been built using the CNN-RNN

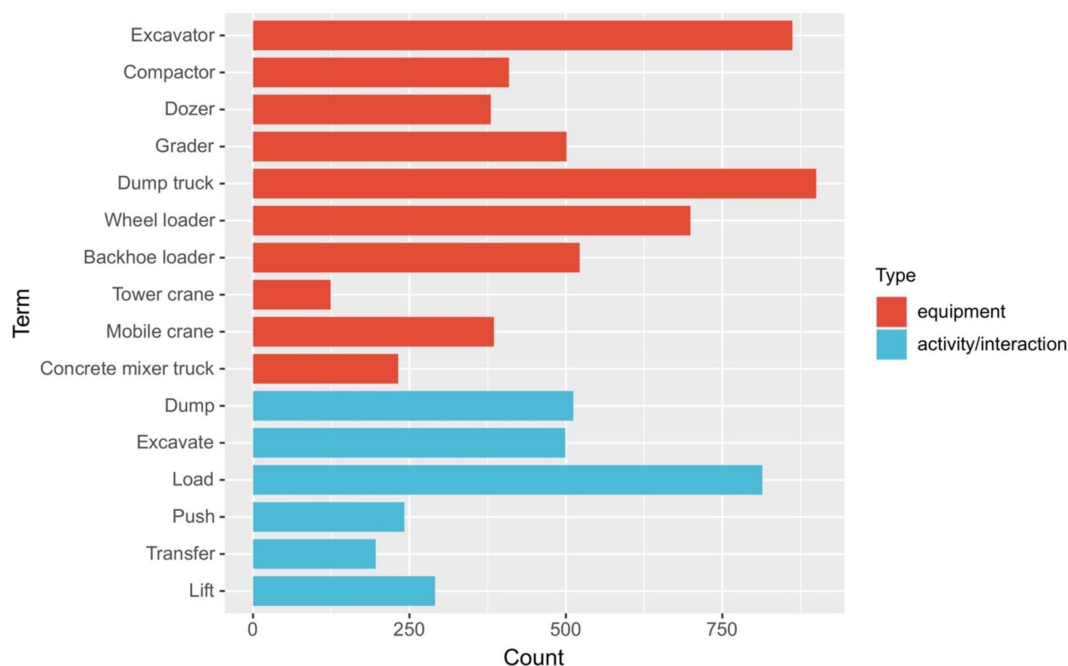
**Fig. 4.** Element distribution of captioning data set.

Table 2. Statistics of top 20 n -grams of the captioning data set

Index	2-gram	Count	3-gram	Count	4-gram	Count
1	Loader is	2,069	Wheel loader is	1,151	A wheel loader is	768
2	Dump truck	1,575	A dump truck	979	A backhoe loader is	615
3	Wheel loader	1,392	A wheel loader	894	A mobile crane is	411
4	Excavator is	1,319	Backhoe loader is	885	Concrete mixer truck is	347
5	On the	1,025	An excavator is	880	Into a dump truck	315
6	An excavator	1,014	A backhoe loader	665	A compactor is compacting	313
7	Backhoe loader	988	A grader is	618	A dump truck is	270
8	A dump	983	Mobile crane is	597	A concrete mixer truck	269
9	Truck is	916	A compactor is	508	An excavator is excavating	258
10	A wheel	898	A mobile crane	468	Wheel loader is loading	243
11	Grader is	896	Dump truck is	445	A grader is grading	213
12	Crane is	761	Compactor is compacting	441	Compactor is compacting the	196
13	Compactor is	722	Concrete mixer truck	401	The wheel loader is	184
14	Mobile crane	715	A dozer is	387	In a construction site	168
15	A backhoe	676	Excavator is excavating	369	An excavator is dumping	166
16	A grader	645	Mixer truck is	350	Waiting to be loaded	266
17	The soil	625	Loader is loading	319	Is traveling on the	163
18	Dozer is	590	The dump truck	318	The soil on the	155
19	Is loading	585	Into a dump	318	Dump truck is waiting	152
20	The road	558	Is traveling on	316	Loading a dump truck	148

approach, and the use of other advanced methods in construction has yet to be investigated. In what follows, we describe in greater detail the selection of the six deep learning image captioning methods from the computer vision community, introduce the five metrics employed for automatic evaluation of these techniques, and give an account of the implementation of the resulting image captioning models.

Method Selection

Baseline Method

The baseline method (Base) consisting of CNN and RNN networks was selected for evaluation. In the baseline method, the ResNet101 network (He et al. 2016) was employed as the encoder and the LSTM network was adopted as the decoder. Fig. 5 shows the architecture of the baseline method. Most studies in construction (Bang and Kim 2020; Liu et al. 2020) have adopted the baseline method for captioning construction images.

Attention Method

In this study, the attention method (Att) was selected as the decoder for testing the construction images (ResNet101 having been selected as the encoder). The attention decoder (Xu et al. 2015) allows neural networks to look at different parts of the image at different steps in the sequence, and this approach has enjoyed considerable success within the computer vision community. The architecture of the attention method is shown in Fig. 6.

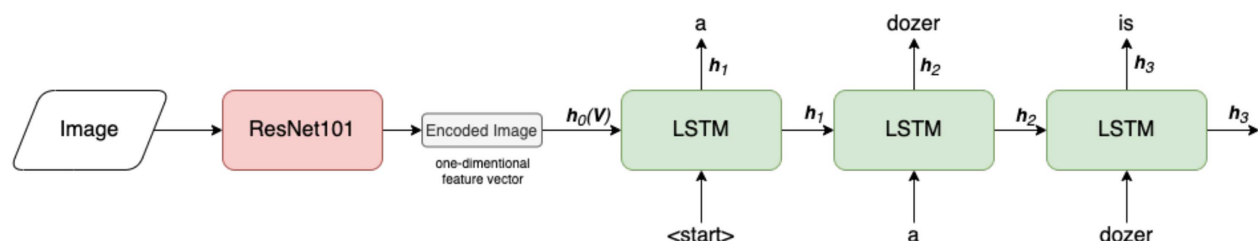
Typically, an attention decoder functions as a small neural network added to an LSTM neural network that takes the hidden state as input (meaning it will look at the sequence generated thus far), and outputs a set of weights for the image feature that indicates what areas the LSTM should focus on (based on a heavy weight). The weights are applied to the image features in order to obtain the feature content; the content is then sent back to the LSTM to help generate the output. In contrast to the baseline method, in the attention method, LSTM functions as an attention network, whereas the encoder networks remain the same.

Transformer Method

The transformer (Tsfm) decoder (Vaswani et al. 2017) is a multi-head attention mechanism that has achieved better performance than the attention decoder in computer vision applications. We implemented the transformer method by integrating the ResNet101 encoder with a transformer decoder. As shown in Fig. 7, the transformer decoder consists of multihead attention layers, normalization layers, and feedforward layers. The multihead attention layers are a set of parallel attention networks that calculate the attention weights, and the feedforward layers (e.g., LSTM) are responsible for conducting the bulk of the decoding work.

Self-Critical Sequence Training

Applying specific training strategies will improve the performance of deep learning image captioning methods. In this regard, the self-critical sequence training (SCST) strategy (Rennie et al. 2017) was integrated in the present study. The SCST adopts reinforcement

**Fig. 5.** Architecture of baseline method.

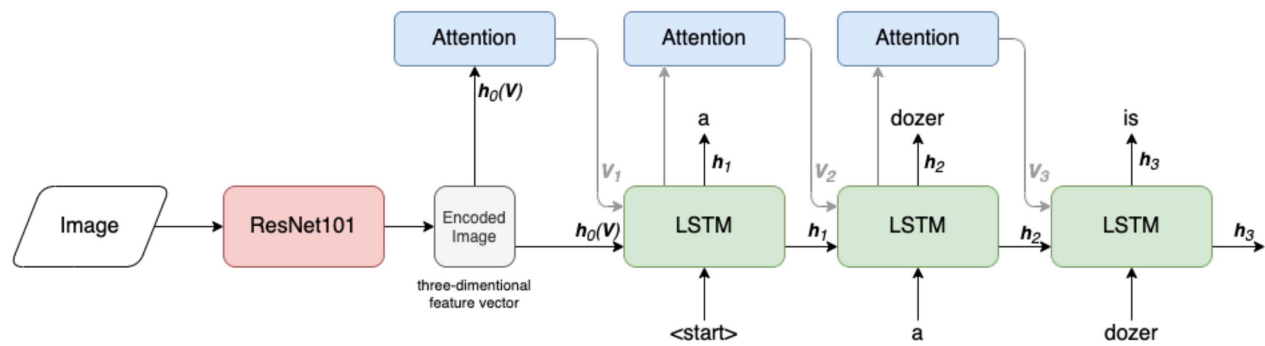


Fig. 6. Architecture of attention method.

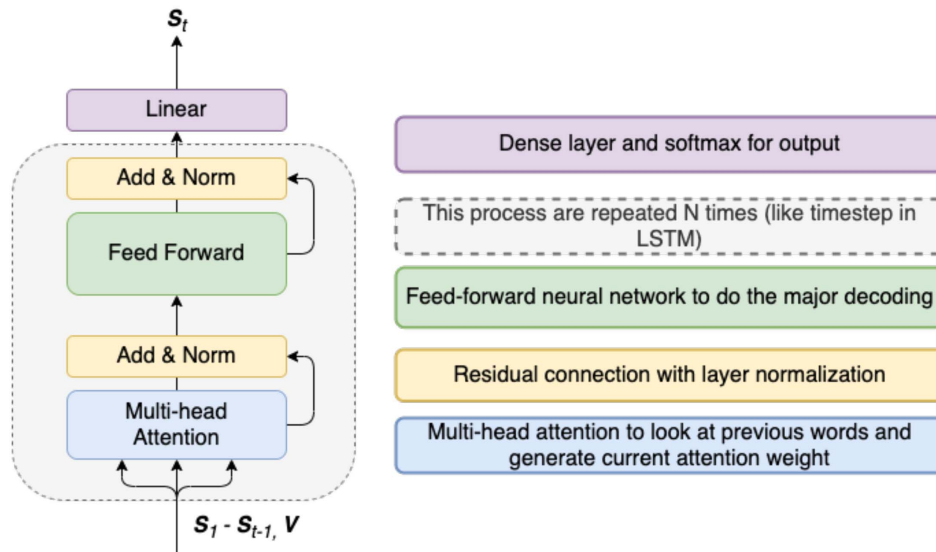


Fig. 7. Illustration of transformer decoder.

learning for the purpose of training deep learning image captioning methods, and it uses a nondifferentiable task metric for optimization. In SCST, two sequences have been estimated in the inference testing procedure, where one is sampled from the softmax distribution and another is greedily sampled. The rewards of two sequences are combined for self-critical as the final loss, which makes SCST more effectively trained in deep learning image captioning. In the present study, the SCST strategy was applied to all three of the aforementioned methods (i.e., Base, Att, and Tsfm).

To sum up, six deep learning image captioning methods [Base, Base-self-critical sequence training (Base-SCST), Att, Att-SCST, Tsfm, and Tsfm-SCST] were selected for testing on the developed captioning data set. In the captioning process, the construction image is firstly processed by the same encoder neural networks ResNet101 (the architecture is described in Table 3) to extract a feature vector with the size of $2,048 \times 1$, and then the extracted feature vector is processed by different decoder neural networks in different image captioning methods to generate natural sentences. Moreover, Table 4 illustrates the technical details of six deep learning image captioning methods in terms of decoder neural networks and the training strategy.

As indicated in Table 4, the number of training epochs was 30 for Base, Att, and Tsfm. For Base-SCST and Att-SCST, the models were first trained for 30 epochs by optimizing the cross-entropy loss and then training for 20 epochs using the SCST strategy.

Table 3. Architecture of ResNet101

Layer name	Output size	Architecture (kernel size and depth)
conv1	$112 \times 112 \times 64$	$7 \times 7, 64$; stride 2
conv2_x	$56 \times 56 \times 64$	3×3 max pool; stride 2
		$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	$28 \times 28 \times 256$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$
conv4_x	$14 \times 14 \times 512$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1,024 \end{bmatrix} \times 23$
conv5_x	$7 \times 7 \times 1,024$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2,048 \end{bmatrix} \times 3$
—	$1 \times 1 \times 2,048$	Average pool

For Tsfm-SCST, the model was trained for 15 epochs for the traditional strategy (optimizing the cross-entropy loss) and for five epochs for the SCST strategy. The training epoch is a hyperparameter determined by fine-tuning in the implementation stage. In this study, the epoch number of Tsfm-SCST was smaller than

Table 4. Technical information of six deep learning image captioning methods

Method	Encoder	Decoder type	Decoder			Training strategy			
			No. of LSTM layers	No. of hidden nodes in LSTM	No. of hidden nodes in Attention	Learning rate	No. of training epochs		
							Cross-entropy loss epochs	SCST epochs	Batch size
Base	ResNet101	LSTM	1	512	N/A	0.0005	30	N/A	10
Base-SCST	ResNet101	LSTM	1	512	N/A	0.00001	30	20	10
Att	ResNet101	LSTM + Attention module	1	512	512	0.0005	30	N/A	10
Att-SCST	ResNet101	LSTM + Attention module	1	512	512	0.00001	30	20	10
Tsfm	ResNet101	Transformer	N/A	N/A	N/A	0.0005	30	N/A	10
Tsfm-SCST	ResNet101	Transformer	N/A	N/A	N/A	0.00001	15	5	10

Note: N/A = not applicable.

Base-SCST and Att-SCST because the authors observed that the Tsfm-SCST converged faster in the training process. This is resulting from the following reasons: (1) Tsfm methods have a more complicated neural network architecture than Base and Att methods; and larger networks are naturally easier to converge; and (2) the SCST increased the converging speed of training larger neural networks.

Evaluation Metric Selection

At present, no single general metric for evaluation of image captioning methods in computer vision is available. For the purpose of this study, five automatic evaluation metrics were adopted in order to assess the performance of deep learning image captioning methods at the sentence level by comparing the ground-truth sentences and the generated sentences. These evaluation metrics were bilingual evaluation understudy (BLEU), recall-oriented understudy for gisting evaluation (ROUGE), metric for evaluation of translation with explicit ordering (METEOR), consensus-based image description evaluation (CIDEr), and semantic propositional image caption evaluation (SPICE). For these metrics, a higher value indicates better captioning performance. The scale of values for CIDEr is 0 to 10, and the scale for the other four metrics is 0 to 1.

BLEU

BLEU (Papineni et al. 2001) measures the overlap between the predicted single word or n -gram (sequence of n adjacent words) and a set of reference sentences. BLEU only measures the word match and sentence length match and does not take the semantic meaning of the words into account. Variations of BLEU include BLEU-1, BLEU-2, BLEU-3, and BLEU-4, where the number appearing after the hyphen signifies the number of words used up to n -grams (the variations listed here are the ones adopted in the present study).

ROUGE

ROUGE (Chin-Yew 2004) uses n -grams to measure the recall score of the generated sentences relative to the reference sentences. The most widely used version of ROUGE, ROUGE-L, was adopted in the present study. ROUGE-L computes the recall and precision of the longest common subsequences between the candidate and reference sentences.

METEOR

METEOR (Lavie and Agarwal 2007) introduces semantic matching for automatic evaluation. It includes lexical match, stemmed word match, synonym match, and paraphrase match. The METEOR score is calculated by mapping the unigrams of the candidate and reference sentences and measuring their alignment with one another.

CIDEr

CIDEr (Vedantam et al. 2014) first converts the words in both the candidate and reference sentences into their root forms and then measures the coexistence frequency of the n -grams in both sentences. The most commonly used version is CIDEr-D because it is capable of preventing outlier scores resulting from poor human judgment. The present study adopted the CIDEr-D version.

SPICE

SPICE (Anderson et al. 2016) calculates the score by measuring the similarity between the scene graph tuples of the candidate and reference sentences. The scene graph includes objects, their attributes, and relationships extracted from the sentence.

Model Implementation

The developed deep learning image captioning models were trained on the developed image captioning training set. All images were resized to 256×256 and normalized based on a mean of (0.485, 0.456, 0.406) and a standard deviation of (0.229, 0.224, 0.225). Moreover, the sentence annotations were tokenized in order to divide them into lists of single words without punctuation. As part of the tokenization process, the <start> and <end> label were added to the beginning and ending of each token list to indicate the start and end of each annotation.

All six deep learning image captioning methods were implemented in the Python language. The encoder (i.e., ResNet101) and decoders (i.e., LSTM, attention, and transformer) adopted in this study were implemented using the Pytorch library. The ResNet101 was pretrained on the ImageNet data set, and the OpenCV library was employed for image input/output. In terms of hardware, the evaluation was conducted on a computer that features two NVIDIA GTX 1080 Ti GPUs (11 GB each), an Intel Core i9-7920X operating at 2.90 Hz CPU with 12 cores, and two 32-GB memory cards. The testing environment used the Ubuntu 16.04 system.

Experimental Evaluation and Results

The trained deep learning image captioning models were validated on the validation set for the purpose of experimental evaluation. Evaluations were carried out at both the sentence level and element level. The sentence-level evaluation followed the standard process employed within the computer vision community for image captioning evaluation, which calculates the similarity between the predicted sentences and the manually labeled sentences. However, the sentence-level evaluation is not intuitive for construction management because these metrics only focus on the holistic similarity among sentences. To address this limitation, the element-level evaluation has been conducted in the experiments, which

Table 5. Results of deep learning image captioning method evaluation

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE
Base	0.587	0.477	0.398	0.320	0.274	0.560	1.499	0.394
Base-SCST	0.546	0.460	0.390	0.318	0.253	0.567	1.549	0.358
Att	0.570	0.463	0.385	0.308	0.267	0.549	1.408	0.382
Att-SCST	0.576	0.484	0.412	0.340	0.274	0.582	1.702	0.418
Tsfm	0.586	0.474	0.392	0.311	0.273	0.556	1.427	0.388
Tsfm-SCST	0.606	0.506	0.427	0.349	0.287	0.585	1.715	0.422

Note: The best performance is indicated by bold.

aimed to evaluate the performance of image captioning methods of retrieving the scene element information (e.g., objects and activities) from construction images.

Sentence-Level Evaluation Results

Table 5 summarizes the sentence-level evaluation results for deep learning image captioning applications in construction. As presented in the table, among the six methods, the Tsfm-SCST was found to achieve the best performance in captioning construction images, attaining the BLEU-1 score of 0.606, BLEU-2 of 0.506, BLEU-3 of 0.427, BLEU-4 of 0.349, METEOR of 0.287, ROUGE of 0.585, CIDEr of 1.715, and SPICE score of 0.422, underscoring the feasibility of the transformer decoder and SCST strategy in deep learning image captioning. In computer vision, the leading scores on the COCO captioning are 0.795 on BLEU-1, 0.635 on BLEU-2, 0.485 on BLEU-3, 0.363 on BLEU-4, 0.573 on ROUGE, 0.277 on METEOR, 1.196 on CIDEr, and 0.213 on SPICE, and these metrics are close to the performance observed in the present study. This indicates that the deep learning image captioning methods under consideration attain comparable results in construction applications to those observed in the computer vision community.

Fig. 8 shows example captioning results generated by six methods in the validation set. It can be seen from these examples that the deep learning image captioning methods under study were capable of correctly describing the contents of construction images in most cases.

The Base method was found in this experiment to rank second in performance, achieving a BLEU-1 score of 0.587, a BLEU-2 of 0.477, a BLEU-3 of 0.398, a BLEU-4 of 0.320, a METEOR of 0.274, a ROUGE of 0.560, a CIDEr of 1.499, and a SPICE score of 0.394, outperforming Att and Tsfm in construction scenarios. In computer vision, in contrast, Att and Tsfm achieved better performance than the Base method. This result demonstrates that the degree of difficulty of image captioning in construction is lower than that in traditional computer vision applications. The implications behind this finding are twofold: (1) currently, the differences of captioning performance among existing methods are insignificant in construction scenarios; and (2) when considering both captioning performance and the processing speed, the Base method is recommended in construction because it achieved comparable performance to the other advanced methods (i.e., Att and Tsfm) while consuming fewer computational resources.

Element-Level Evaluation Results

Image captioning provides a holistic solution for understanding the scene information (i.e., objects, activities, and relationships between objects) in an image. By analyzing the generated sentences, this scene information can be retrieved for various uses related to construction object and activity recognition. To validate the feasibility of deep learning image captioning methods for construction scene analysis, the Tsfm-SCST method was evaluated at the

element level in terms of its ability to recognize machine objects and activities in images.

As with the sentence-level evaluation, in the element-level evaluation, the Tsfm-SCST method was trained on the training set and validated on the validation set. In the evaluation process, first, the machine objects and activities were extracted from the generated sentences. Then, the retrieved elements were compared with the elements appearing in the ground-truth sentences. In the present study, the precision, recall, and F1 score were used for element-level evaluation. The calculation of these metrics is illustrated in Eqs. (1)–(3)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

where TP = true positive, which is the number of machine objects appearing in both the generated sentences and the ground-truth sentences; FP = false positive, which is the number machine objects appearing in the generated sentences but not appearing in the ground-truth sentences; and FN = false negative, which is the number of machine objects appearing in the ground-truth sentences but not in the generated sentences.

Table 6 presents the element-level evaluation results for the Tsfm-SCST method in terms of its ability to recognize machine objects. As can be seen in the table, Tsfm-SCST was found to achieve, on average, a precision of 91.1%, recall of 83.3%, and an F1 score of 86.6% for the validation set, meaning that it has a close but slightly lower performance than the state-of-the-art object-detection methods in construction scenarios. The Tsfm-SCST method achieved its highest precision (100%) in recognizing tower crane objects and its highest recall (92.9%) in recognizing grader objects.

Table 7 summarizes the element-level evaluation results produced by Tsfm-SCST in terms of recognizing machine activities. This research is limited in scope to the three activities appearing most frequently in the captioning data set: loading, dumping, and excavating. As indicated in the table, the Tsfm-SCST achieved an average precision of 42.5%, recall of 45.4%, and F1 score of 41.2% in terms of construction machine activity recognition.

There are two reasons why the deep learning image captioning performed relatively poorly in construction activity recognition:

- The limited number of training data. In the proposed study, a captioning data set that contains 4,000 construction images and 8,226 captions has been developed for training and validation, which is relatively small compared with the data sets in computer vision. When the number of training data increases, the performance of recognizing construction activities will be improved; and



Base: a backhoe loader is pushing soil

Base-SCST: a backhoe loader is travelling on the road

Att: a backhoe loader is loading the soil from the field

Att-SCST: a backhoe loader is loading soil

Tsfm: a backhoe loader is loading the soil from the field

Tsfm-SCST: a backhoe loader is loading the soil



Base: a compactor is compacting road

Base-SCST: a compactor is compacting the road

Att: compactors in UNK compacting asphalt

Att-SCST: a compactor is compacting the road

Tsfm: a compactor is compacting the road

Tsfm-SCST: two compactors are compacting on the site



Base: a concrete mixer truck is moving on the road

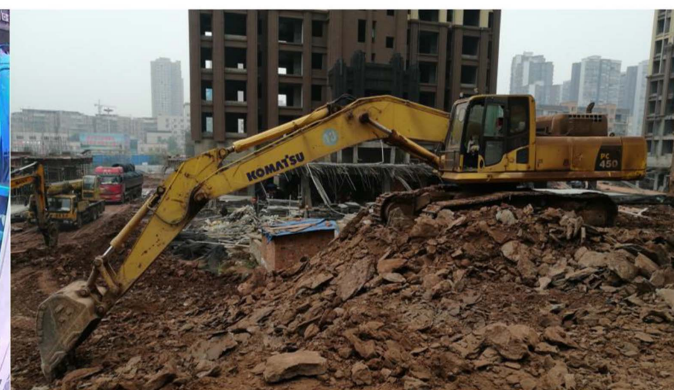
Base-SCST: a concrete mixer truck is travelling on the road

Att: a concrete mixer truck is parked on the side of a road

Att-SCST: a concrete mixer truck is parked on the site

Tsfm: a concrete mixer truck is waiting at the construction site

Tsfm-SCST: a concrete mixer truck is leaving the site



Base: an excavator is excavating soil

Base-SCST: an excavator is pushing soil

Att: an excavator is excavating soil at a site

Att-SCST: an excavator is excavating soil

Tsfm: an excavator is excavating the soil into the dump truck

Tsfm-SCST: an excavator is excavating the soil from the construction site

Fig. 8. Example captioning results in the validation set. (Images by authors.)

Table 6. Element-level evaluation results for Tsfm-SCST for machine object recognition

Machine objects	Precision (%)	Recall (%)	F1 score (%)
Excavator	87.6	89.9	88.7
Compactor	92.9	90.8	91.9
Dozer	94.6	76.8	84.8
Grader	92.9	92.9	92.9
Dump truck	79.6	89.9	84.4
Concrete mixer truck	91.9	61.8	73.9
Wheel loader	93.1	77.7	84.7
Backhoe loader	90.9	85.7	88.2
Tower crane	100.0	82.1	90.2
Mobile crane	87.0	85.7	86.3
Average	91.1	83.3	86.6

Table 7. Element-level evaluation results of Tsfm-SCST for machine activity recognition

Machine activities	Precision (%)	Recall (%)	F1 score (%)
Load	40.0	23.6	29.7
Dump	38.7	74.0	50.8
Excavate	48.7	38.5	43.0
Average	42.5	45.4	41.2

- The limited types of construction activities included. Currently, we only considered few types (less than five) of activities for each construction machine and many of activities are overlapped, which increases the difficulty of using deep learning image captioning for construction activity recognition.

The aforementioned results suggest that further development is needed in future works before deep learning image captioning can be considered as a replacement method for activity recognition in construction scenarios. Image captioning is developing fast in computer vision community, and the authors believe this technique will satisfy the requirements of construction management in the close future. When more powerful image captioning models are developed, our proposed method will help researchers in our field to quickly adopt new image captioning models for construction management applications.

Discussions

The experimental results indicated the successful achievements of the objectives underlying this research. The specific research findings and limitations are discussed as follows:

- For construction scene analysis, image captioning methods have two advantages over existing construction methods (performing object detection, activity recognition separately and combining all information together), which are (1) a faster inference speed; specifically, the inference speed of image captioning methods is around 30 frames per second (fps), whereas the inference speed of existing scene analysis methods is slower than 1 fps (Kim et al. 2016); and (2) a simpler training process. The image captioning methods only require one annotated data set and can retrieve multiple scene information from the captioning results, whereas existing scene analysis methods require multiple data sets for training object detection and activity recognition. However, this does not mean image captioning is more effective than object detection and activity recognition for vision-based monitoring in construction. In fact, they are three different tasks and cannot be compared directly. Meanwhile, object detection and activity recognition cannot be replaced by image captioning because they can provide clear localization and activity information of specific construction objects that cannot be provided by image captioning.
- In construction monitoring, the practical implications of image captioning include automated image documentation, evaluation of infrastructure damages, and efficient image querying. For image documentation, image captioning methods can generate natural sentences from images/videos to describe what happened in construction sites. Currently, project managers still need to manually prepare daily reports. Additionally, image captioning has the potential to automatically document the project progress and safety issues in the text format, which will reduce manual documentation efforts. Moreover, image captioning can be used to evaluate infrastructure damages because the generated

sentences contain explanatory information. Image captioning can generate an explanatory sentence to identify the damage type and level from infrastructure images. For example, Chun et al. (2021) adopted image captioning for evaluating the damages of bridges. Meanwhile, image captioning can be employed for efficient image querying in construction management. A large number of images has been accumulated in construction projects, and the generated sentences from image captioning can serve the role of text-index, which is able to provide versatile information of individual construction images.

- The SCST strategy was shown to improve the performance of image captioning methods in construction. In the evaluations, the three methods in the present study adopting SCST obtained an average BLEU-1 of 0.576, BLEU-2 of 0.483, BLEU-3 of 0.410, BLEU-4 of 0.336, METEOR of 0.271, ROUGE of 0.578, CIDEr of 1.655, and a SPICE of 0.399. For the methods not adopting SCST, the average performance was 0.581 for BLEU-1, 0.471 for BLEU-2, 0.392 for BLEU-3, 0.313 for BLEU-4, 0.271 for METEOR, 0.555 for ROUGE, 1.445 for CIDEr, and 0.388 for SPICE. With the exception of BLEU-1 and METEOR, the methods adopting SCST were found to outperform the methods without SCST. The results indicate that applying specific strategies in training can improve the performance of image captioning methods.
- Fig. 9 shows some example captioning errors produced by the Tsfm-SCST method. The top three most frequent captioning errors encountered in this study are (1) misrecognition of primary machines (e.g., in Failure 3, the primary machine wheel loader was misidentified as a grader); (2) misrecognition of activities (e.g., in Failure 4, the wheel loader was driving on the road, whereas it was misidentified as being engaged in dumping soil); and (3) misrecognition of working contents (e.g., in Failure 7, the Tsfm-SCST failed to recognize the working contents and yielded the unreasonable sentence “a mobile crane is lifting the construction site”).

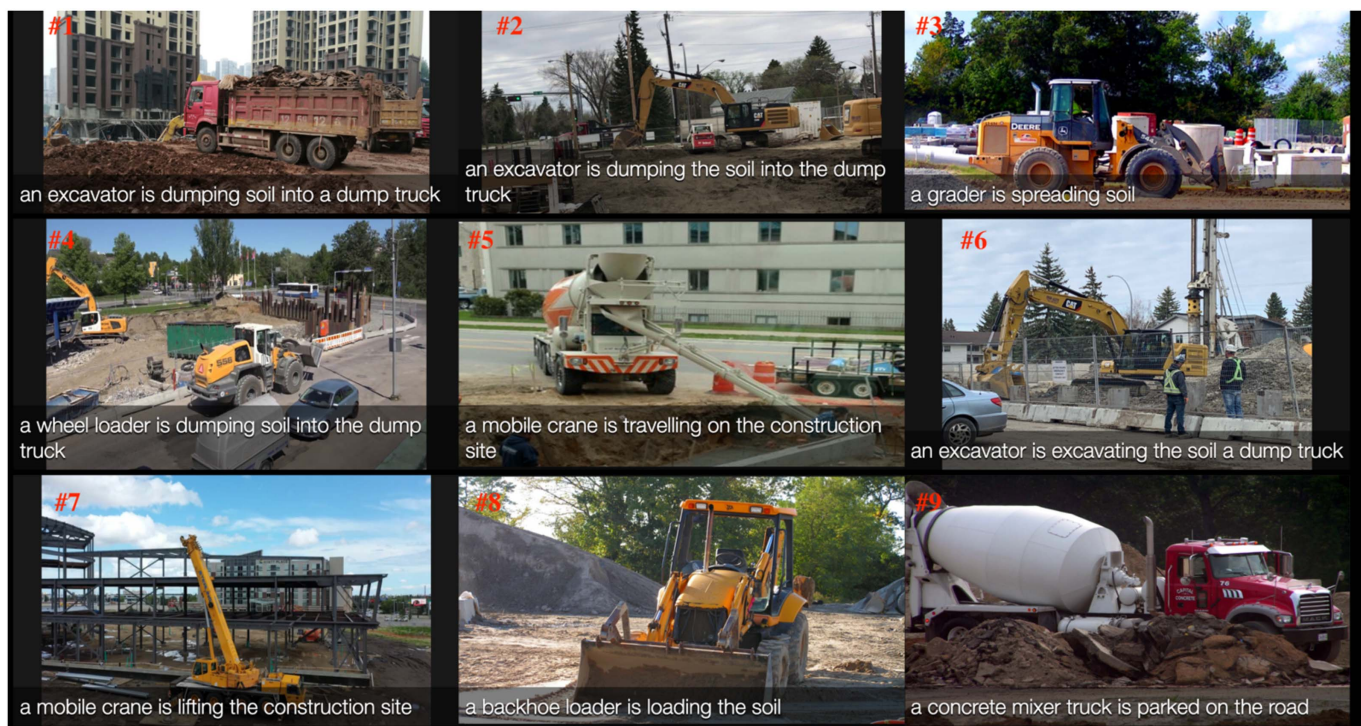


Fig. 9. Example captioning errors committed by Tsfm-SCST. (Images by authors.)

- There are three main reasons of the captioning errors: (1) the limited number of training images restricts the learning ability of image captioning methods, which results in the miscategorization of objects and activities; (2) the limited types of construction activities included in the captioning data sets increases the captioning difficulty of construction activities; and (3) the existing image captioning methods are built upon the encoder-decoder mechanism, which focuses on the holistic image features instead of fully considering the sequence relationships of the construction elements in the captioning process.
- The present research has three limitations that need to be addressed in future research: (1) the quantity of the captioning image data set needs to be increased. The present research developed a captioning image data set containing 4,000 images and 8,226 captioning sentences. Compared with data sets in computer vision, the quantity of the proposed data set was relatively small; (2) the present research only investigates encoder-decoder-based image captioning methods, whereas there are some deep learning image captioning methods available built upon other mechanisms [e.g., captioning by detection (Johnson et al. 2015; Lu et al. 2018)] whose application to construction warrants investigation in future work; and (3) when using image captioning for machine object recognition, the captioning method can only retrieve the primary and cooperating machine objects, and small machine objects in the background are not captured. Object-detection methods, on the other hand, are capable of recognizing all machine objects appearing in the given images or videos.

Conclusions and Future Works

In this research, we conducted a feasibility study to investigate the performance of deep learning image captioning in construction scenarios. We proposed a linguistic schema to deconstruct construction machine images into primary objects, cooperating objects, activities, working contents, and supplementary information. Using the linguistic schema, professional descriptions can be annotated for the purpose of training deep learning image captioning methods. A captioning data set was developed containing 4,000 images and 8,226 sentences. In turn, six deep learning image captioning methods were tested on the captioning data set in the sentence-level evaluation. In the element-level evaluation, the Tsfm-SCST method achieved F1 scores of 86.6% and 41.2% for recognizing scene objects and activities, respectively, in construction images.

The contributions of this research are threefold. First, a three-step method has been proposed for integrating deep learning image captioning methods in construction. Using the proposed method, the feasibility of image captioning methods in construction can be investigated. Second, a linguistic schema for annotating construction machine images has been developed. Using this schema, an annotated image data set has been developed for training deep learning methods for the purpose of captioning images featuring construction machines. Third, six deep learning image captioning methods have been compared using automatic metrics in the computer vision community. Meanwhile, an analysis of the efficacy of various image captioning methods in identifying construction objects and activities has been conducted using the Tsfm-SCST method, demonstrating the potential of applying image captioning in scene element analysis.

Future works will focus on expanding the quantity of the captioning image data set. Currently, only 4,000 images in the ACID data set are annotated, with an average of two ground-truth sentences for each image. In the future, all 10,000 images in the ACID data set will be annotated, with five ground-truth sentences per image.

Meanwhile, combining image captioning methods with advanced training strategies (e.g., SCST and data augmentation) in construction scenarios is another important future work. By adopting these strategies, the performance of image captioning can be significantly improved when the number of training data is limited. We will also work on integrating image captioning methods into advanced construction applications, such as a construction image querying system and autogeneration of daily reports.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request (e.g., captioning data sets and models).

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